

Humberto Sandmann



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Computer scientist and educator with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. With more than a decade of university teaching alongside senior industry practice in software architecture, machine learning, and high-performance computing, I bring research depth and applied engineering experience into the classroom.

Research Interests

DeFi, Smart contracts and decentralized applications

Small-world networks and complex systems

Computation in large-scale systems

Artificial neural networks and deep learning

Nonlinear systems and network dynamics

Synchronization of coupled oscillators

Education

mar.2012

↑

mar.2007

D.Sc.

thesis

[networks](#)

Polytechnic School of the University of Sao Paulo

[Spike patterns and computation in dynamical neural](#)

doi

[10.11606/T.3.2012.tde-05092012-165022](#)

advisor

[Prof. Dr. Emilio Del Moral Hernandez](#)

presentation

[slides](#)

supported by

[CNPq](#) - grant 140632/2007-7

(mar.2007 → dec.2009) $\wedge$ (sep.2010 → feb.2011)



sep.2010
 ↑
 aug.2009

D.Sc. Internship [Max Planck Gesellschaft für Dynamik und Selbstorganisation](#)
 advisor [Prof. Dr. Marc Timme](#)
 supported by [CAPES - grant BEX 4076-09-5](#)
 [jan.2010 → aug.2010](#)
 [Pró-Reitoria da Pós-Graduação da USP](#)
 [aug.2009 ^ dec.2009 \(flight tickets GRU ⇌ FRA\)](#)
 [José Humberto Sandmann - my father](#)
 [aug.2009 → dec.2009 \(German language classes/exams\)](#)
 [Max Planck Institute - Networks Dynamics Group](#)
 [aug.2009 → nov.2009 \(accommodation in Goettingen\)](#)



jun.2006
 ↑
 feb.2003

M.Sc. [Polytechnic School of the University of Sao Paulo](#)
 dissertation [Prediction of time series using architecture based on neuro-fuzzy systems](#)
 doi [10.11606/D.3.2006.tde-01042009-095125](#)
 advisor [Prof. Dr. Marco Túlio Carvalho de Andrade](#)



dec.2002
 ↑
 feb.1999

B.Sc. [Faculdade de Engenharia Industrial](#)
 [Computer Science Bachelor](#)

dec.1998
 ↑
 feb.1996

[Escola Técnica Jorge Street](#)
 Technician in Industrial Computation

Academic Appointments & Teaching

now
 ↑
 jun.2018

[Insper](#)
 Sao Paulo, Brazil

now
 ↑
 feb.2020

Associate Professor

Platform, Microservices and APIs	Spring Cloud, Spring Boot, PostgreSQL, Redis, RabbitMQ, Kafka, Docker, K8s	2026.2, 2026.1, 2025.2, 2025.1, 2024.1
Neural Networks and Deep Learning	Artificial Neural Networks, Perceptron, Multi Layer Perceptron, Convolutional	2026.2, 2025.2

		Neural Networks, GAN, VAE, CLIP, Stable Diffusion	
	Machine Learning	PCA, t-SNE, UMAP, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Classification, Regression, Clustering, EDA, Pre- processing, Feature Engineering	2026.2, 2026.1
	Coding Interviews	Course to prepare for coding interviews.	2026.1
	TecWeb	JS, React	2025.2
	Cloud Computing	MAAS , Juju , OpenStack , K8s , IaaS , PaaS , Computer Networks	2025.1, 2024.2
	Design of Software	Python	2026.2, 2025.0, 2024.1, 2023.2, 2022.1, 2021.2, 2021.1
	Information System for Administration and Economy	Excel, VBA, Python, NumPy, Pandas, Stochastic Processes, RoI	2023.2, 2023.1, 2022.2
	Spreadsheet Automation in Excel	Excel, VBA	2021.1 , 2020.1
jan.2020 ↑ aug.2018	Assistant Professor		
	Design of Software	Python	2019.2, 2019.1
	Foundations for Physical	Python, Fourier Transform, Wavelet	2018.2

		Communication	Transform, Signal Processing	
jun.2026	ESPM			
↑	Sao Paulo, Brazil			
jun.2014	Associate Professor			
	Machine Learning	Machine Learning, PCA, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, Page Rank, PySpark, Classification, Regression, Clustering, Pre-processing, Feature Engineering		2025.2
	Applied Mathematics	Propositional Logic, Boolean Algebra, Set Theory, Polynomials, Functions, Sequences, Series, Logarithm, Combinatorial Analysis, Limits, Derivatives, Linear Algebra, Matrix Algebra, Graph Theory		2025.1, 2024.1, 2023.2, 2023.1, 2022.2, 2022.1, 2021.2, 2021.1, 2020.2, 2020.1
	Oriented Object Programming	Java, Design Patterns, Docker, Spring Cloud, Microservices		2025.1, 2024.1, 2023.1, 2022.1, 2021.1
	Statistics	Statistics, Probability, Random Variables, Distributions, Sampling, Estimation, Hypothesis Testing, Regression Analysis		2023.2
	Software Quality	Software Quality, Software Testing, Software Architecture, Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation, Continuous Integration, Continuous Delivery		2024.2, 2022.2

	Architecture and Operating Systems	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	ESPM 16-bits 2021.1, 2020.2, 2020.1, 2019.2, 2019.1, 2018.2, 2018.1, 2017.1, 2016.2, 2016.1, 2015.2, 2015.1, 2014.2
	Artificial Intelligence and Society	Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs, Transformers, LSTMs, BERT, Ethics, Bias	2022.1 (IR), 2021.1 , 2020.2 , 2020.1 , 2019.2
	Introduction for Artificial Intelligence	Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs	2020.2 , 2020.1 , 2019.2 , 2019.1
apr.2015 ↑	Toledo Prudente Centro Universitário Presidente Prudente, Brazil		
mar.2015	Visiting Professor		
	MBA: Big Data and Machine Learning	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr, R, Octave, Self-Organizing Maps, Bayesian Decision Trees, Multilayer Perceptron	2015.1
dec.2015 ↑	FIAP Sao Paulo, Brazil		
mar.2012	Associate Professor		
	Distributed systems	Sockets, RMI, SOAP, Microservices, J2EE, Design Patterns	2015
	MBA: Big Data and No-SQL Modelling	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr	2014.2
	Data Structure and Algorithm II	C, Data Structure, Algorithm, Complexity, Linked List, Stack, Queue, Tree, Red-Black Tree, AVL Tree, B-Tree, Hash Table, Graph, Dijkstra Algorithm, Prim Algorithm	2013

Algorithms	C, Programming, Algorithm, Complexity	2013, 2012.2
Database I and II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Object- Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence, Synonym	2013, 2012
Programming III	Pointers, C, C++, Object-Oriented Programming, Inheritance, Polymorphism, Encapsulation, Abstract Class, Interface, Exception Handling	2012

jul.2013

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mar.2012

[Universidade Cruzeiro do Sul](#)

Sao Paulo, Brazil

Associate Professor

Mobile Programming	Android	2012.2
Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	2012.1
Software Engineering	Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation	2012.1
Introduction to Programming	C, Programming, Algorithm	2012.1
Database II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence	2013.1, 2012.1

jul.2011	Universidade Nove de Julho		
↑	Sao Paulo, Brazil		
mar.2011	Temporary Professor		
	Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	2011.1
	Introduction to Programming	C, Programming, Algorithm	2011.1
	Web Programming	HTML, CSS, JavaScript	2011.1
jul.2011	Polytechnic School of the University of Sao Paulo		
↑	Sao Paulo, Brazil		
mar.2008	Teaching Assistant		
	Experimental Pattern Recognition, Modeling and Neurocomputing	Logic Gates, Linear Regression, Pattern Classification, UC Irvine Machine Learning Repository , NIST Data Gateway , Multilayer Perceptron, EEG Signals, PCA	2011.1*
	Modeling for Signal Processing: Neural Networks and Learning	Artificial Neural Networks, Multilayer Perceptron, Backpropagation, Forecasting, Time Series, Classification	2009.1 , 2008.1*
	Laboratory of the Foundations of Computing Engineering	Computer Architecture, Assembly Language, Memory Management, Process Management, Register, Stack, Turing Machine, Finite State Machines, Von Neumann Architecture, Java, Orientated Object Programming	2008.1†

* grants from CAPES (PAE)

† grants from FAPESP

Professional Experience

mar.2025 ↑ feb.2006	H-Sandmann Sao Paulo Area, Brazil Founder	Machine Learning, Software Architecture.
oct.2023 ↑ mar.2012	SelsanTech Sao Paulo Area, Brazil Co-Founder	Machine Learning, Software Architecture.
jun.2019 ↑ may.2018	RockSpoon Sao Paulo Area, Brazil - Palo Alto, United States Senior Software Architect and Machine Learning Engineer	POS, ERP, Payroll, JavaEE, Postgres, Software Architecture, OpenCV, Machine Learning, Computer Vision, GoLang, Android, Kotlin.
jul.2016 ↑ oct.2015	Banco Votorantim Sao Paulo Area, Brazil Contractor: Senior Consultant for Computational Optimization	RoE, Banking Business, High-Performance Computing, J2EE, Spring Boot, Automating Testing, Jenkins.
nov.2011 ↑ aug.2011	TecBan Sao Paulo Area, Brazil Contractor: Senior Consultant for Computational Optimization	ATM, J2EE, InMemory Database, High-Performance Computing.
jun.2007 ↑ feb.2006	ABN Amro Bank Sao Paulo, Brazil Contractor: Senior Software Engineer/Architect	Stock Market, Trading, Core Banking, Software Architecture, Cryptography, HSM, BouncyCastle, PKI, PCKS, Java, Servlet, Database.
jan.2006 ↑ aug.2005	Politec Sao Paulo, Brazil Senior System Analyst	J2EE, Software Quality, Software Architecture, Banking Business.
jul.2005 ↑ jul.2004	EverSystems Sao Paulo, Brazil Senior Software Engineer/Architect - BankBoston - CitiBank - Elly Lilly	PKI, BouncyCastle, Cryptography, HSM, Core Banking, Stock Market, Java, Servlet, Database, Communication Protocol, Synchronization and Replication.

mar.2003	Central Online	Object-Oriented Programming, LMS .
↑	Sao Paulo, Brazil	
sep.2002	Senior System Analyst	
sep.2002	Datamace	Payroll, ERP, Software Architecture.
↑	Sao Paulo Area, Brazil	
oct.1999	Senior System Analyst	
oct.1999	Heraeus Electro-Nite	Compliance, ERP, PCP.
↑	Sao Paulo Area, Brazil	
mar.1998	Computer Programmer	

Publications

Selected peer-reviewed publications. Full and up-to-date list: [ORCID 0000-0001-7758-695X](#).

E. Del-Moral-Hernandez; H. Sandmann; G. Araujo. **Context dependent pattern recognition — A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information**. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), Atlanta, GA, USA, 2009, pp. 663–670. IEEE. doi: [10.1109/IJCNN.2009.5179061](#).

L.A. da Silva; H. Sandmann; E. Del-Moral-Hernandez. **A self-organizing architecture of recursive elements for continuous learning**. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), Hong Kong, China, 2008 (IEEE World Congress on Computational Intelligence). IEEE. doi: [10.1109/IJCNN.2008.4634190](#).

E. Del-Moral-Hernandez; H. Sandmann; L.A. da Silva. **Pattern recovery in networks of recursive processing elements with continuous learning**. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), Budapest, Hungary, 2004. IEEE. doi: [10.1109/IJCNN.2004.1379877](#).

Grants & Funding

CNPq	Doctoral fellowship — grant 140632/2007-7	mar.2007 → dec.2009 $\wedge$ sep.2010 → feb.2011
CAPES	Doctoral internship abroad (Max Planck Institute) — grant BEX 4076-09-5	jan.2010 → aug.2010
CAPES (PAE)	Teaching assistantship fellowship — Polytechnic School, USP	2008, 2011

FAPESP

Teaching assistantship fellowship – Polytechnic
School, USP

2008

Academic Service & Memberships

IEEE – Institute of Electrical and Electronics Engineers

Member #60216475

- **Computational Intelligence Society (CIS)** – member since 2004; Brazilian committee member (apr.2009 → aug.2009 ∧ oct.2010 → mar.2012)
- **Signal Processing Society (SPS)** – 2005 → 2009
- **Engineering in Medicine and Biology Society (EMBS)** – 2005 → 2009

INNS – International Neural Network Society

Student member #3951

Conference Service

Program Committee member and reviewer for the **International Joint Conference on Neural Networks (IJCNN)** – the flagship conference of the **INNS** and the IEEE Computational Intelligence Society.

IJCNN 2020	Reviewer	Glasgow, Scotland	
IJCNN 2019	Program Committee & Reviewer	Budapest, Hungary	program
IJCNN 2018	Reviewer	Rio de Janeiro, Brazil	
IJCNN 2017	Program Committee & Reviewer	Anchorage, AK, USA	program
IJCNN 2015	Reviewer	Killarney, Ireland	program

Languages

Portuguese	native
English	advanced
German	B1

 This CV website was designed and built with the assistance of AI — Anthropic's Claude. All content is curated and verified by Humberto Sandmann.